LIFTING MAPS THROUGH POWERS OF A PRINCIPAL IDEAL MODULO THE A-POWER TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof quotients by the torsion submodule, verifies completeness of the quotient, uses the exact rows to view C as a fibre product, lifts to the quotient by the nonzerodivisor lemma, and reassembles the map via the fibre product description

Editorial note. The label lemma-get-morphism-principal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the lifting lemma for principal ideals with nonzerodivisor (lemma-get-morphism-nonzerodivisor of the same chapter)
- completeness of quotients: C/C[a^infty] is I-adically complete when the intersection of C[a^infty] + a^nC equals C[a^infty] (Algebra, Lemma algebra-lemma-quotient-complete)
- the fibre product description of C from the exact sequence with the torsion submodule
- existence of the integer d with C[a^infty] intersect a^dC = 0: for Noetherian C the torsion stabilizes at C[a^e] and Artin-Rees gives the intersection bound (Algebra, Lemma algebra-lemma-Artin-Rees)

Source reference. Original lemma 00IN: sources/stacks-chow/tags.txt, source lines 685–685. Prerequisite pointer invoked at proof line(s) 1309; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 031A: sources/stacks-chow/tags.txt, source lines 3977–3977. Prerequisite pointer invoked at proof line(s) 1319; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Prerequisite pointer invoked at proof line(s) 1296; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AK6: sources/stacks-restricted/pinned-original-source.tex, source lines 1242–1242. Prerequisite pointer invoked at proof line(s)

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

1335; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJQ: sources/stacks-restricted/pinned-original-source.tex, source lines 395–395. Prerequisite pointer invoked at proof line(s) 1314; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $I = (a)$ be a principal ideal of a Noetherian ring A. Let B be an object of (0AL4). Assume given an integer $c \geq 0$ such that $\mathrm{Ext}^1_B(NL^\wedge_{B/A}, N)$ is annihilated by a^c for all B-modules N. Let C be an I-adically complete A-algebra. Assume given an integer $d \geq 0$ such that $C[a^\infty] \cap a^d C = 0$. Let $n > \max(2c, c+d)$. For any A-algebra map $\psi_n : B \to C/a^nC$ there exists an A-algebra map $\varphi : B \to C$ such that $\psi_n \bmod a^{n-c} = \varphi \bmod a^{n-c}$.*

Proof. Let $C \to C'$ be the quotient of C by $C[a^\infty]$. The A-algebra C' is I-adically complete by Algebra, Lemma 031A and the fact that $\bigcap(C[a^\infty] + a^nC) = C[a^\infty]$ because for $n \geq d$ the sum $C[a^\infty] + a^nC$ is direct. For $m \geq d$ the diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & C[a^\infty] & \longrightarrow & C & \longrightarrow & C' & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & C[a^\infty] & \longrightarrow & C/a^mC & \longrightarrow & C'/a^mC' & \longrightarrow & 0 \end{array}$$

has exact rows. Thus C is the fibre product of C' and C/a^mC over C'/a^mC' for all $m \geq d$. By Lemma 0AK6 we can choose a homomorphism $\varphi' : B \to C'$ such that φ' and ψ_n agree as maps into $C'/a^{n-c}C'$. We obtain a homomorphism $(\varphi', \psi_n \bmod a^{n-c}C) : B \to C' \times_{C'/a^{n-c}C'} C/a^{n-c}C$. Since $n - c \geq d$ this is the same thing as a homomorphism $\varphi : B \to C$. This finishes the proof. $\square$